c-only slope law: the Q/R-ratio design of the c slot

1 Definitions

Where c enters (0928_obs_c_only_v1, §1): the two Jacobian entries

$$\bar{a}'(\hat{h};\, c) = \frac{1-c}{\hat{h}^2}, \qquad M[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k], \qquad F_e(4,5) = -\frac{M[k]}{\hat{h}^2}, \qquad H_2(5) = \frac{\nabla_d\bar{w}_d[k]}{\hat{h}^2}$$

Model of the c slot (random walk; $Q_{55} = 0$ is the base arm)

$$c[k+1] = c[k] + n_c[k], \qquad \text{Var}\; n_c[k] = Q_{55}[k]$$

Kalman recursion of the slot, information form (I_k = this step's information on c)

$$P_{55}^{-}[k] = P_{55}^{+}[k-1] + Q_{55}[k], \qquad \frac{1}{P_{55}^{+}[k]} = \frac{1}{P_{55}^{-}[k]} + I_k, \qquad I_k = \frac{1}{P_{55}^{+}[k]} - \frac{1}{P_{55}^{-}[k]} \;\geq\; 0$$

One reading of c this step: its variance R_k; the Q/R ratio ρ_k; the memory N_k

$$R_k = \frac{1}{I_k}, \qquad \rho_k = \frac{Q_{55}[k]}{R_k}, \qquad N_k = \frac{1}{\sqrt{\rho_k}}$$

Steady state for a given ρ (R_k frozen): the variance of an N-reading average

$$\frac{1}{P_{55}} = \frac{1}{P_{55} + R_k\rho} + \frac{1}{R_k} \quad\Longrightarrow\quad P_{55} \simeq R_k\sqrt{\rho} = \frac{R_k}{N} \qquad (\rho \ll 1)$$

Identity used by the information-driven forgetting (ib3): s_k is I_k in units of P_{55}^{+}

$$s_k = 1 - \frac{P_{55}^{+}[k]}{P_{55}^{-}[k]} = I_k\, P_{55}^{+}[k]$$

2 The design

Hold the steady-state P_{55} at one number P_0: the ratio follows

$$\rho_k = \left(\frac{P_0}{R_k}\right)^2 \quad \Longrightarrow \quad Q_{55}[k] = R_k\,\rho_k = \frac{P_0^2}{R_k} = P_0^2\,I_{k-1}$$

$$P_{55} \simeq R_k\sqrt{\rho_k} = P_0, \qquad N_k = \frac{R_k}{P_0}$$

$$I_k \to 0 \text{ (hold, far field)} \implies Q_{55} \to 0,\ N_k \to \infty$$

$$P_0 = P_{55}[0] \text{ (default)}, \qquad \text{scan } \sqrt{P_0} \in \{0.10,\ 0.20,\ 0.30\}$$

The three arms in one table (same slot, same identity)

arm	$Q_{55}[k]$	ρ_k	P_{55} steady
base	0	0	shrinks only
ratio	$P_0^2\,I_{k-1}$	$(P_0\,I_k)^2$	P_0
ib3, $\lambda_k = \max(e^{-\beta s_k}, 0.99)$	$(\lambda_k^{-1} - 1)\,P_{55}^{+} \simeq \beta\,I_k\,(P_{55}^{+})^2$	$\beta\,(I_k\,P_{55}^{+})^2$	clamp: $1/(100\,I_k)$

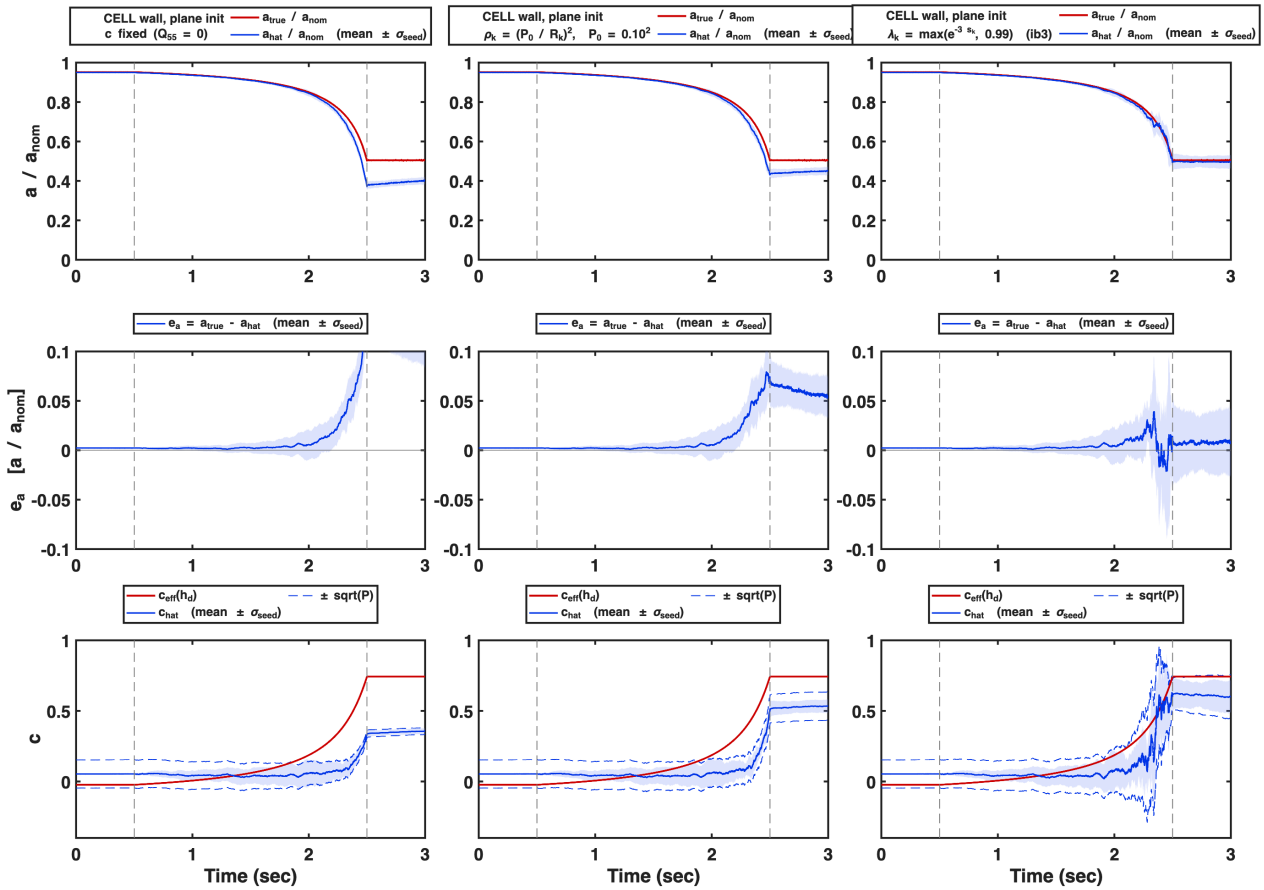
What $\hat{c}$ is compared with: c_{eff} (local truth) and c^* (zeroes $e_{\bar{a}}$ at the wall)

$$c_{\text{eff}}(h) = 1 - h^2\,\bar{a}'_{\text{true}}(h), \qquad \Delta s_j^c = \frac{|M_j|}{\hat{h}_j^2}, \qquad e_{\bar{a}}(\text{wall}) = \sum_{j \in \text{descent}} \left(c_{\text{eff},j} - \hat{c}_j\right)\Delta s_j^c$$

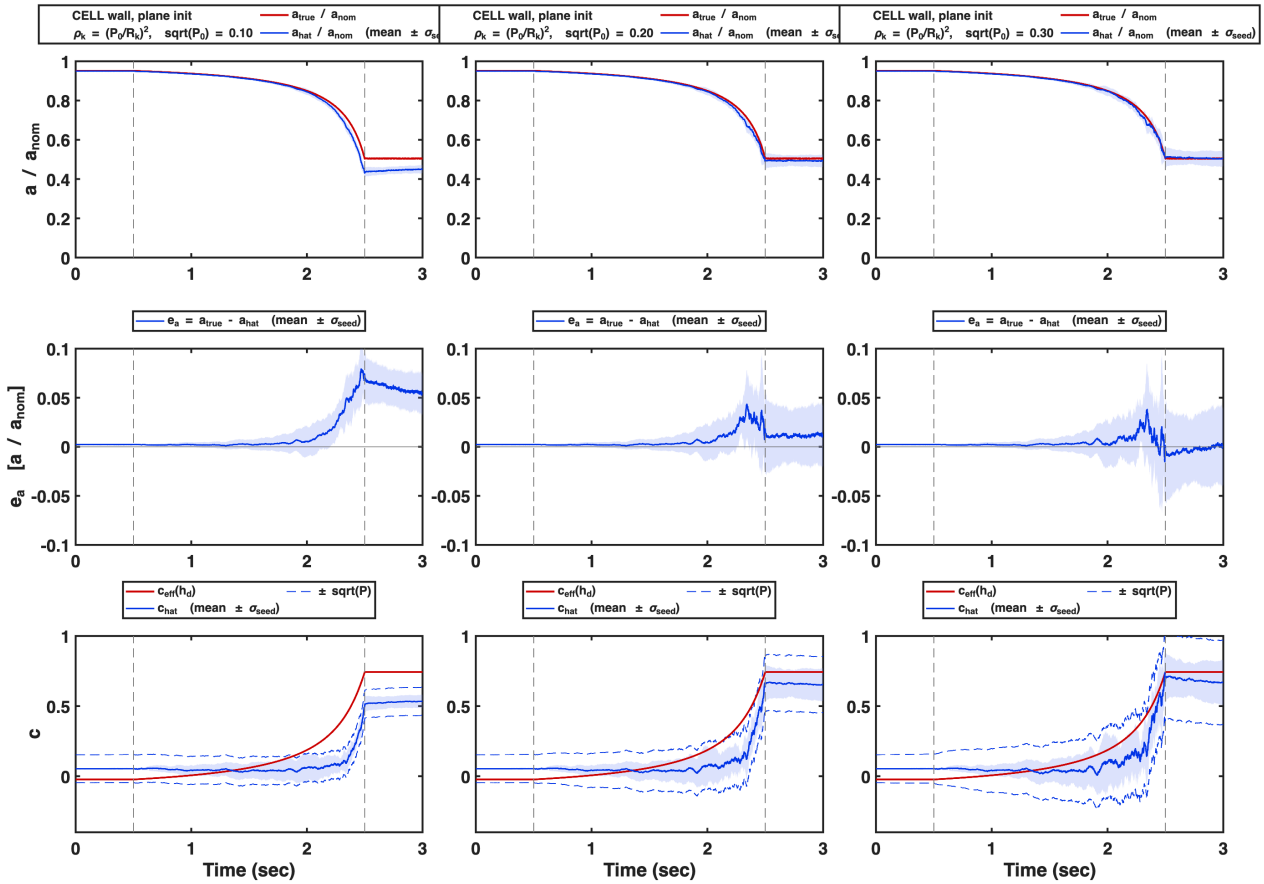
$$\hat{c} \equiv c^* \iff e_{\bar{a}}(\text{wall}) = 0, \qquad c^* = \frac{\sum_j \Delta s_j^c\, c_{\text{eff},j}}{\sum_j \Delta s_j^c}$$

c^* : cell, canon $22.2 \to 1.1\,R$: 0.483; plane: 0.002

3 Cell wall, plane init: base | ratio $\sqrt{P_0} = 0.10$ | ib3

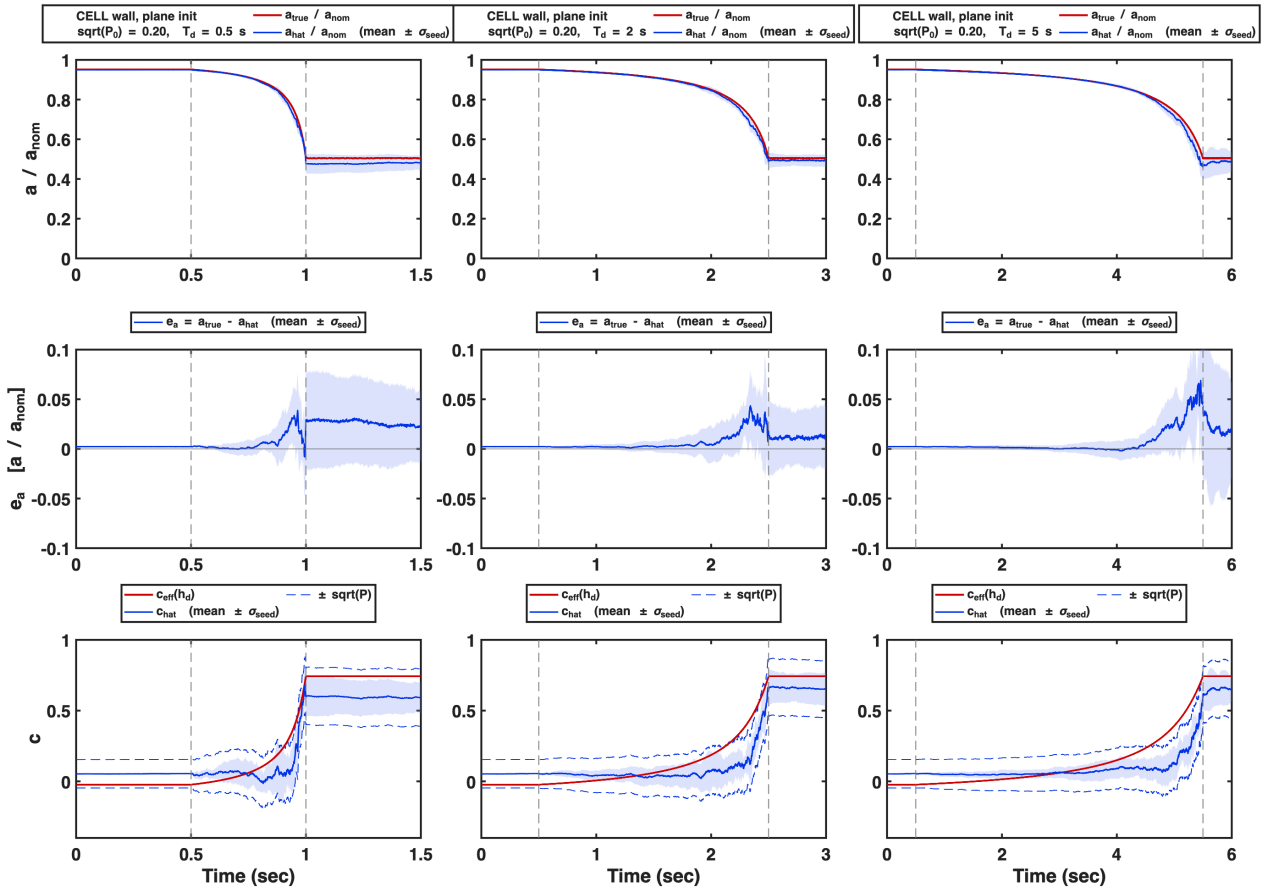


4 Cell wall, plane init: the P_0 scan, $\sqrt{P_0} = 0.10 \mid 0.20 \mid 0.30$

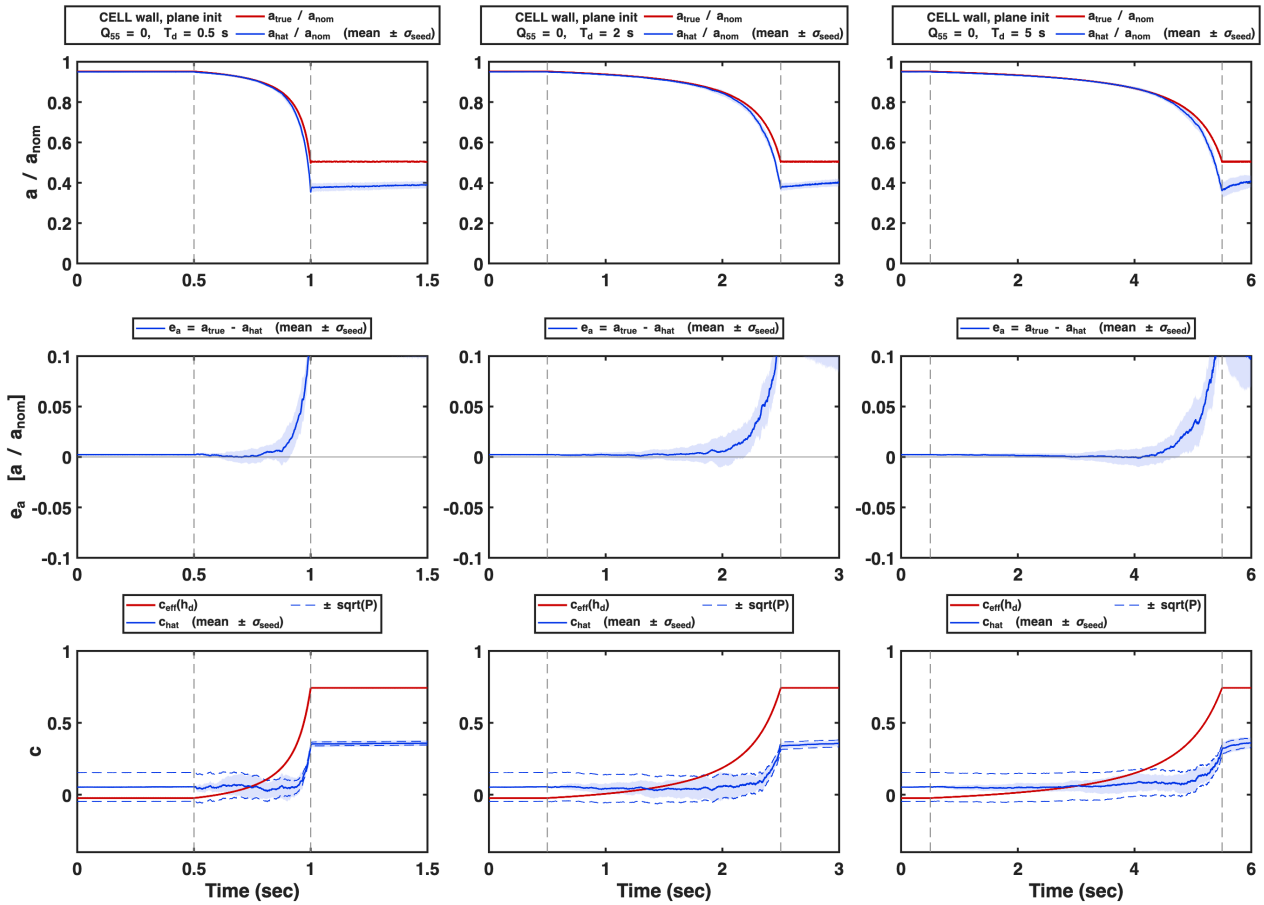


5 Cell wall, plane init, $\sqrt{P_0} = 0.20$: three descent speeds

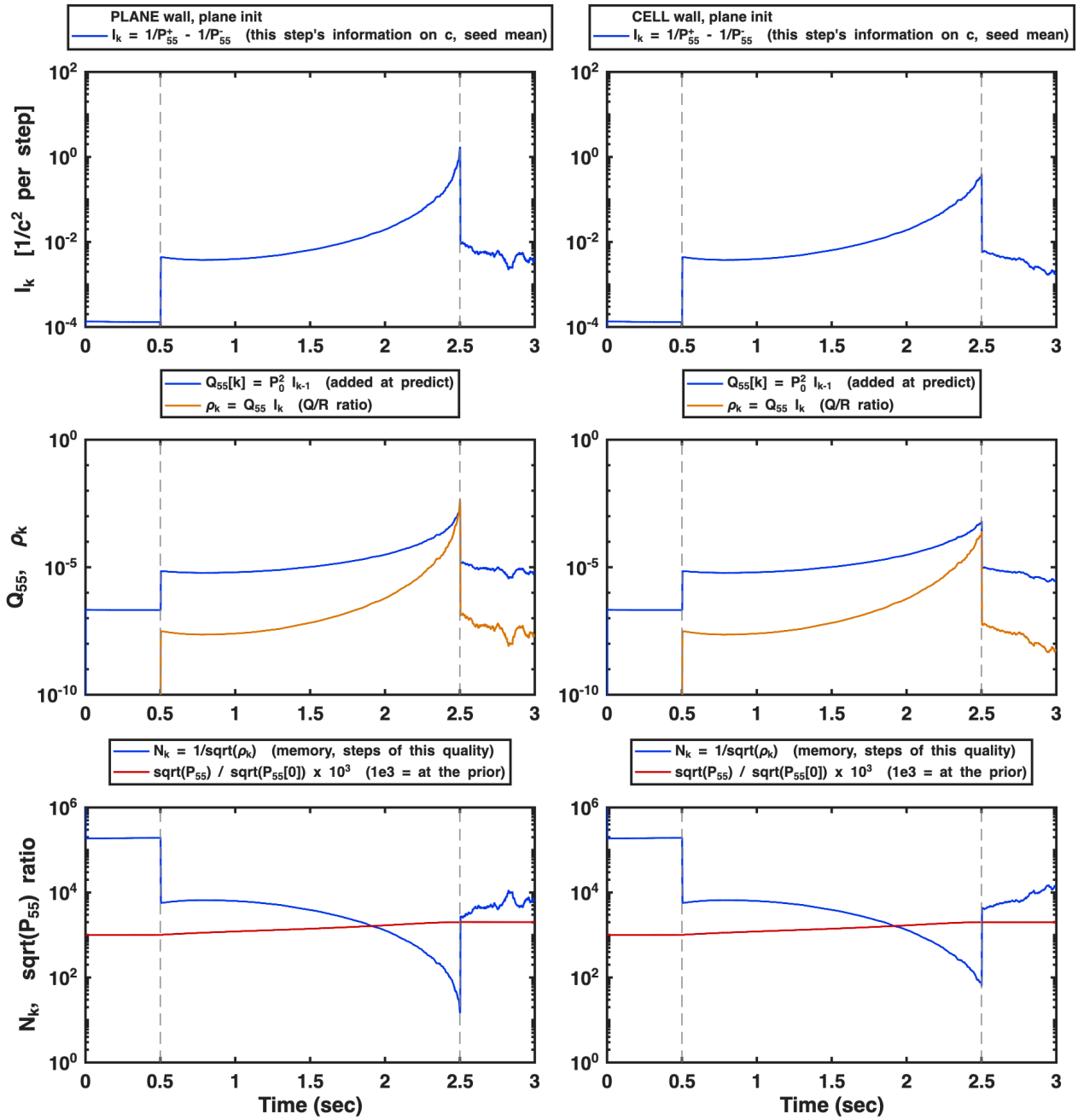
$$T_d = 0.5 \mid 2 \mid 5 \text{ s}$$



6 Cell wall, plane init, $Q_{55} = 0$ (base): the same three speeds

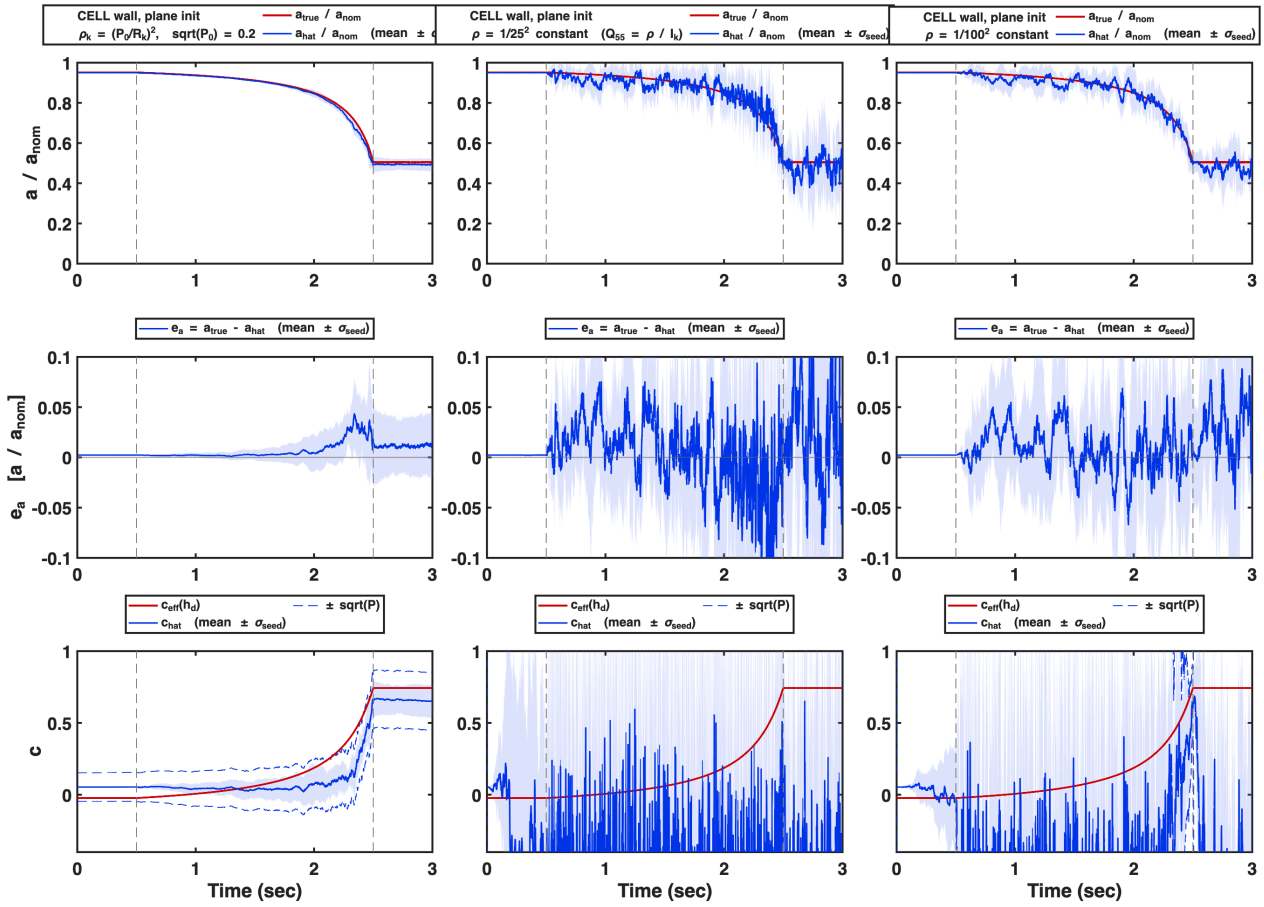


7 Plane | cell, ratio $\sqrt{P_0} = 0.20$: I_k, Q_{55}, ρ_k, N_k, $\sqrt{P_{55}}$ along the run



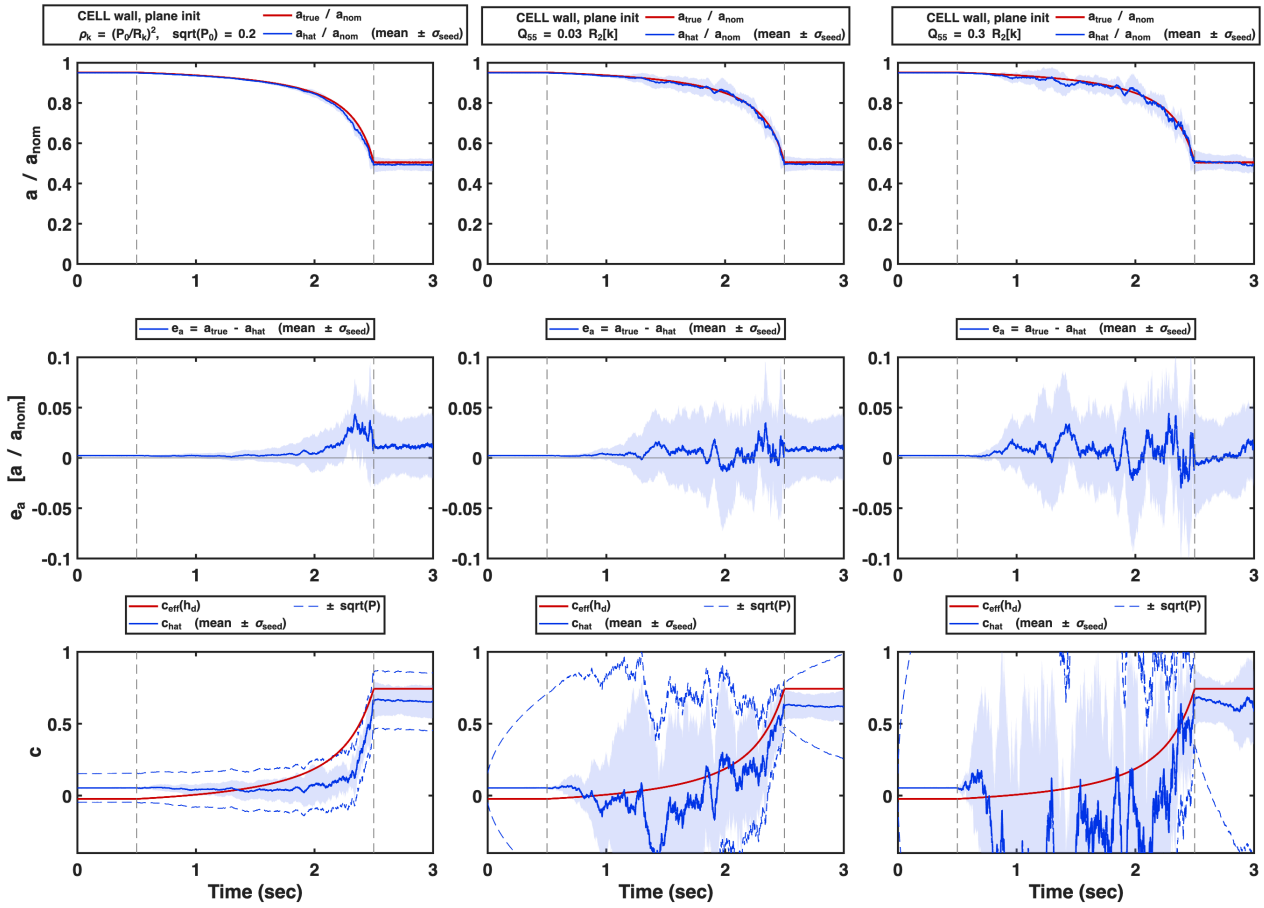
8 The opponents (cell, ramp 2 s): ratio $\sqrt{P_0} = 0.2$ | constant $\rho = 1/25^2$ | constant $\rho = 1/100^2$

$$Q_{55} = \frac{\rho}{I_k}, \qquad P_{55}^{\mathrm{eq}} = \frac{\sqrt{\rho}}{I_k} \;\to\; \infty \;(I_k \to 0)$$



9 The opponents (cell, ramp 2 s): ratio | $Q_{55} = 0.03\,R_2$ | $Q_{55} = 0.3\,R_2$

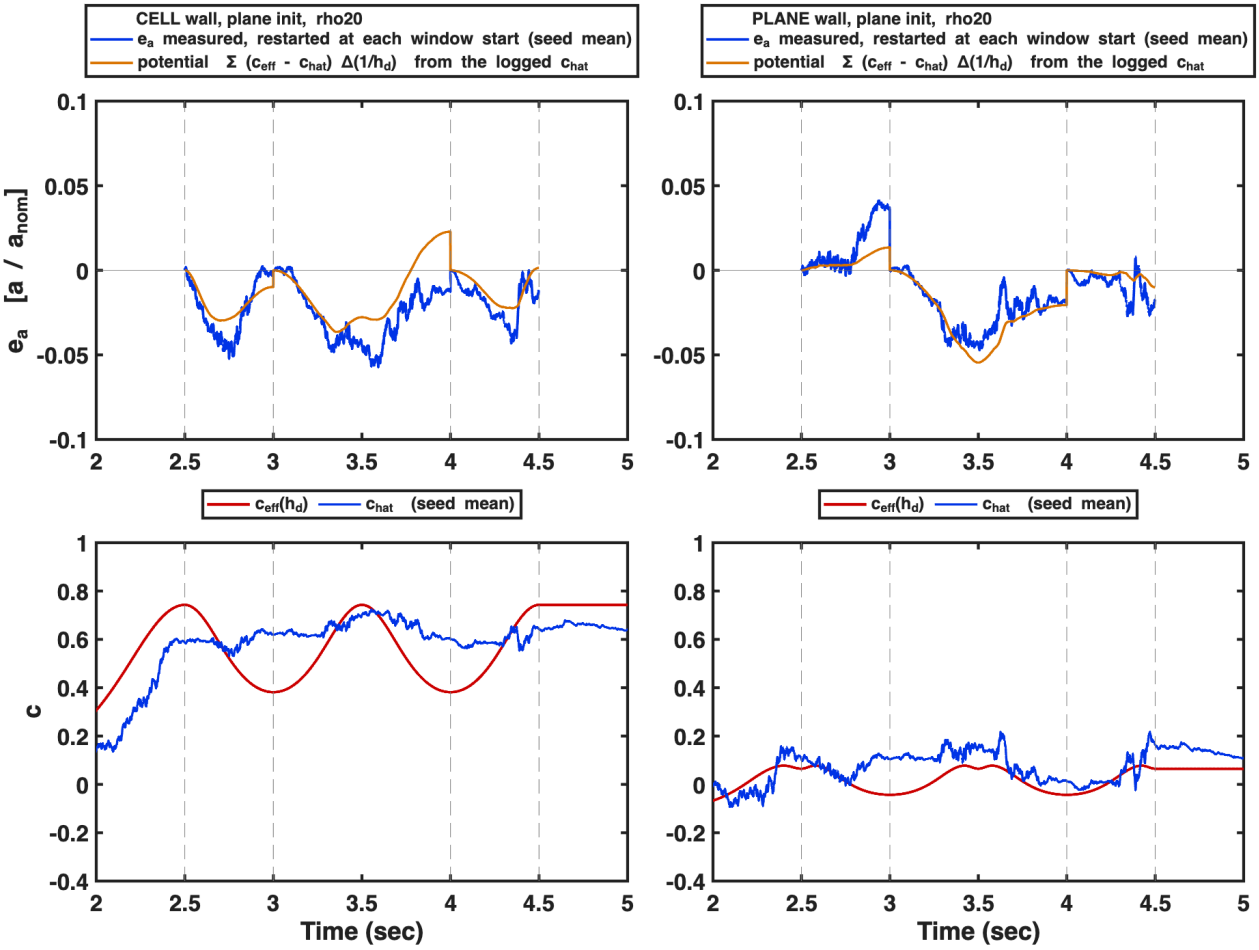
$Q_{55} = \rho\, R_2[k], \qquad P_{55}^{\mathrm{eq}} = \sqrt{\dfrac{\rho R_2[k]}{I_k}}$ (R_2 large where I_k small: far field, hold)



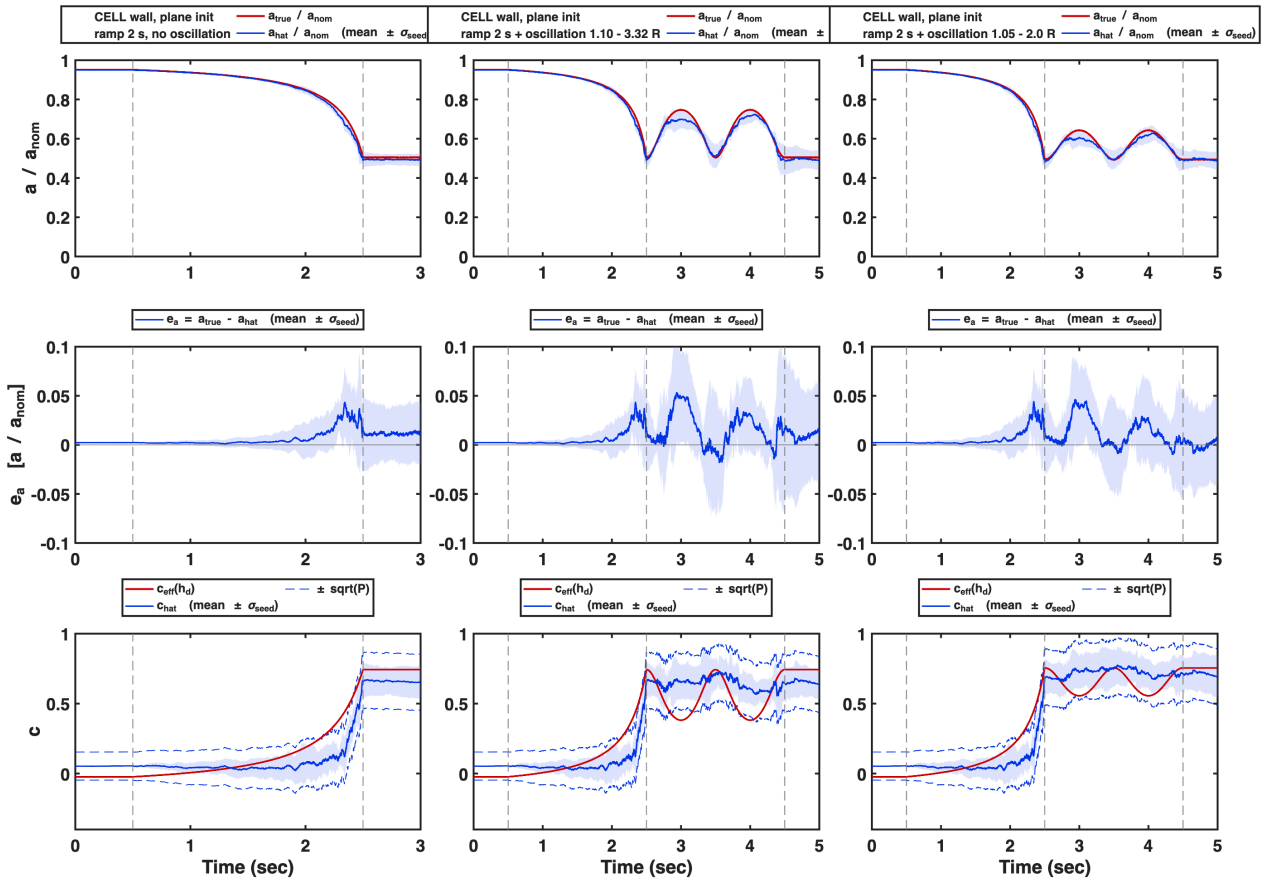
10 What $\hat{c}$ is compared with on the oscillation: the open-loop potential of a constant c

$$e_{\bar{a}}(h) = \int_{h_{\mathrm{top}}}^{h} \left(c_{\mathrm{eff}}(h') - \hat{c}\right) \frac{dh'}{h'^2}, \qquad c^*_{\mathrm{band}} = \frac{\int_{\mathrm{band}} c_{\mathrm{eff}}\, dh/h^2}{\int_{\mathrm{band}} dh/h^2}$$

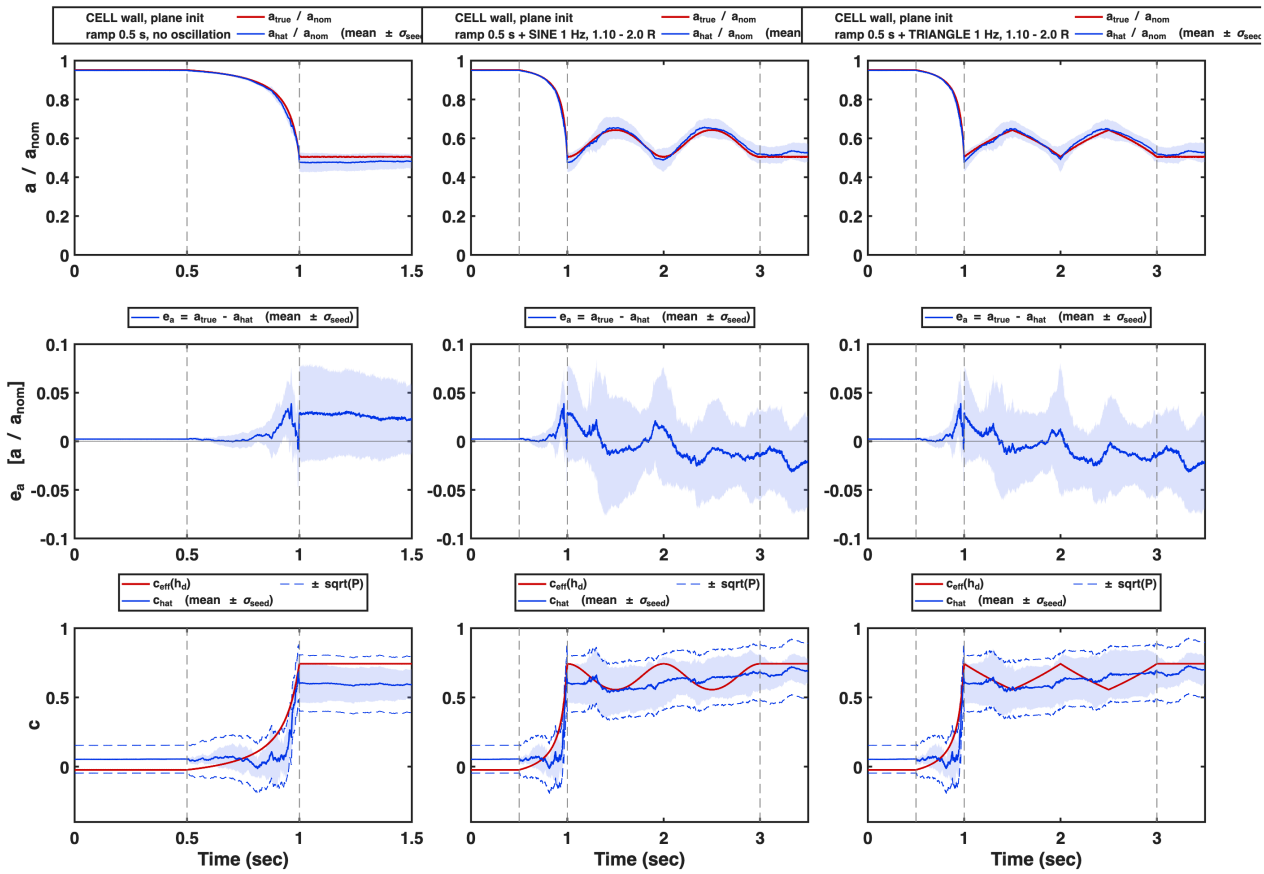
1.10–3.32 R : $c^*_{\mathrm{band}} = 0.60$, in-band extreme 0.026; 1.05–2.0 R : 0.67, 0.011



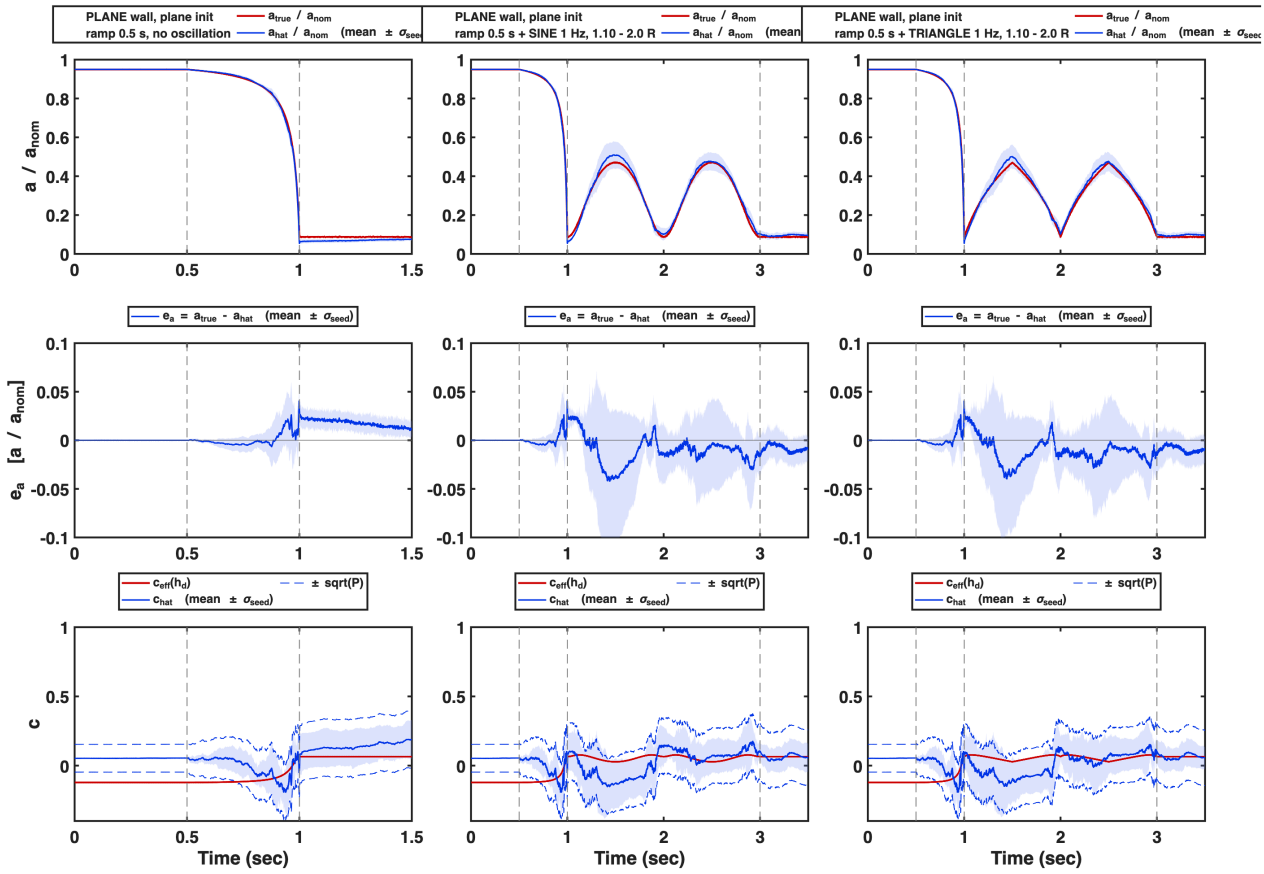
11 Cell, ratio $\sqrt{P_0} = 0.2$: ramp 2 s | ramp 2 s + sine 1.10–3.32 R | ramp 2 s + sine 1.05–2.0 R (constant-velocity descent)



12 Cell, ratio $\sqrt{P_0} = 0.2$, 0.5 s descent + 1 Hz 1.10–2.0 R: no oscillation | sine | triangle

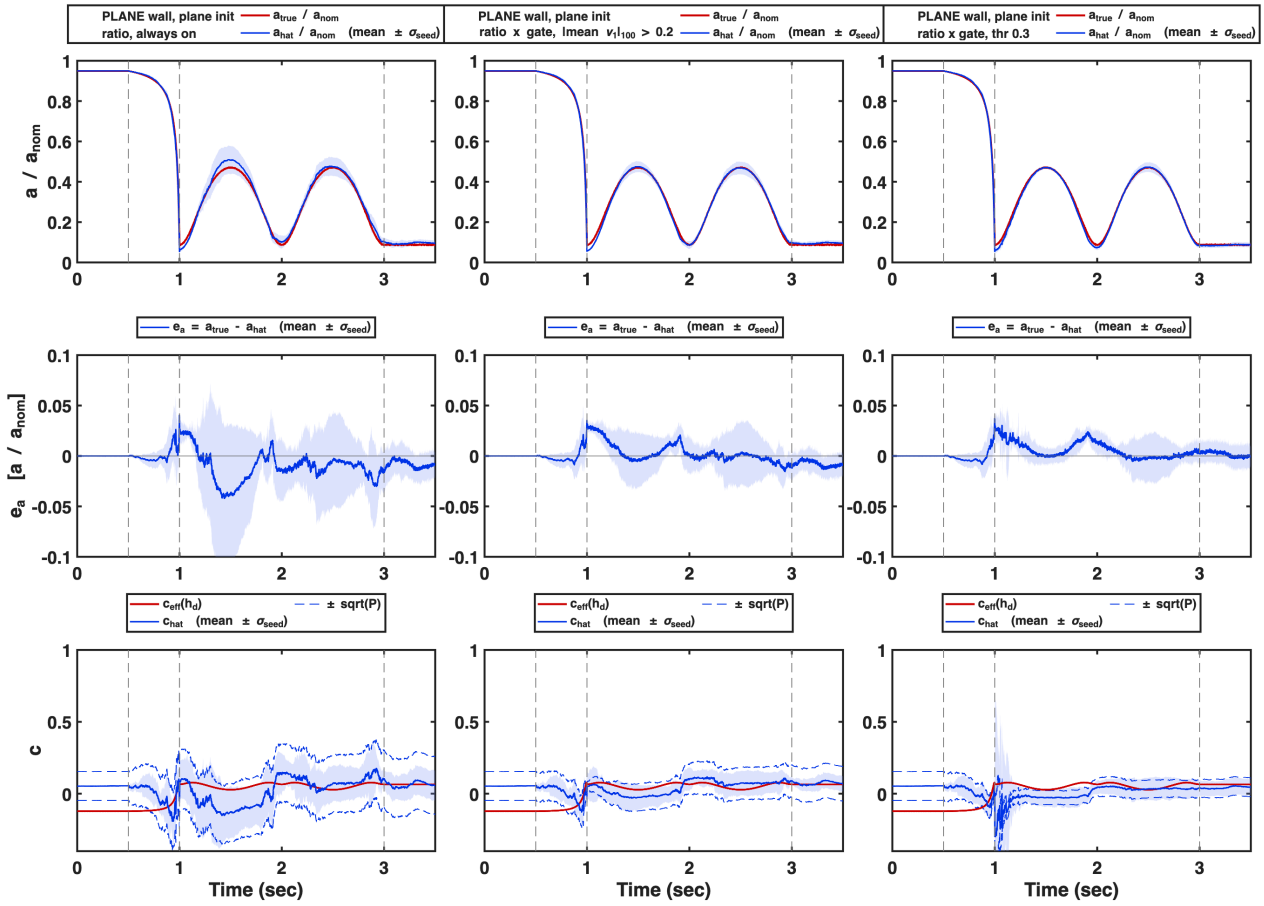


13 Plane, same three arms (the cost side)



14 The innovation gate (plane): ratio | ratio × gate thr 0.2 | thr 0.3

$$\nu_1[k] = \frac{\mathrm{innov}_1[k]}{\sqrt{S_1[k]}}, \qquad s[k] = \frac{1}{100}\sum_{i=0}^{99} \nu_1[k-i], \qquad Q_{55}[k+1] = P_0^2\, I_k \mathbb{1}\big[\,|s[k]| > \mathrm{thr}\,\big]$$



15 Three designs on one trajectory (cell): ratio | ratio × gate | constant R_2 ratio

